

Medical Robotics

Prof. Marilena Vendittelli

Sapienza Università di Roma

Lecture 9

GVF-based Control of Cooperative Systems for Retinal Surgery and Impedance-based Dynamic Virtual Fixtures

2nd Semester, Academic Year 2025–2026

23 February – 29 May 2026 · 60 hours · 6 ECTS

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1. Lecture Overview

This lecture develops two detailed case studies in virtual fixture (VF) control, building directly on the general framework introduced in Lecture 8. The first case study is the **Johns Hopkins University Steady-Hand Robot (SHR)**, a cooperative system for retinal microsurgery, which uses an admittance-based guidance VF combined with a virtual Remote Centre of Motion (RCM) constraint. The formal GVF control law — including the treatment of reference targets and reference paths — is derived and its convergence properties are discussed. The second case study moves to **teleoperated systems** and introduces **impedance-based dynamic virtual fixtures**, applied to the da Vinci Research Kit, where the GVF is enforced as a spring-damper force overlaid on the master tool manipulator. The lecture concludes with a brief survey of similar robotic platforms for ophthalmic surgery developed at other institutions.

Connection to previous lectures. Lecture 7 introduced the Acrobot as an instance of a *forbidden-region* VF in a cooperative context. This lecture addresses the complementary case: *guidance* VFs, where the fixture attracts the tool toward a desired path or target rather than repelling it from a forbidden region. The mathematical structure — an anisotropic admittance or impedance relating force to motion — is the same; only the geometric definition of the preferred subspace changes.

2. Clinical Motivation: Retinal Vein Cannulation

Several retinal pathologies — including retinal vein occlusion — could benefit from the delivery of anticoagulant or thrombolytic drugs directly into a vein on the surface of the retina (**retinal vein cannulation**). The procedure requires inserting a needle into a vessel whose outer diameter is on the order of **100 μm** . This is smaller than the root mean square amplitude of physiological hand tremor in a surgeon, which has been measured at **182 μm** . The procedure is therefore not safe to perform freehand with current clinical standards, but would have great utility if it could be executed reliably.

The fundamental challenge is one of scale: at the microscopic level, the surgeon's own hand motion is the dominant source of positioning error. Two complementary strategies address this:

1. A **cooperative robotic system** that shares control of the surgical tool with the surgeon and filters physiological tremor, and
2. **Software-generated guidance virtual fixtures** that further constrain the tool motion to remain on the planned path.

3. The Steady-Hand Robot (SHR)

3.1. System Description and Required Degrees of Freedom

The **Johns Hopkins University Steady-Hand Robot** was designed to cooperatively share control of a surgical tool with the surgeon while satisfying the performance, accuracy, and safety requirements of retinal microsurgery. The control modality is the same as the Acrobot: the surgeon and the robot hold the tool simultaneously, and the robot modulates its apparent impedance to filter unwanted motion while obeying the surgeon's intent.

The degrees of freedom required for eye surgery decompose as follows:

- **Approach:** at least 3 DOFs to position the system relative to the patient.
- **Insertion:** 3 DOFs (one translation along the tool axis, two rotations).

- **Retinal surgery:** 4 DOFs for executing the actual procedure (three rotations of the tool orientation, one translation for insertion/retraction of the tool).

An additional kinematic constraint must be satisfied: the **Remote Centre of Motion (RCM)** at the scleral entry point. As with all intra-ocular instruments, the tool must pivot about the point where it enters the eye to avoid lacerating the sclera. In the SHR, this RCM is enforced *virtually* through a software constraint in the control law, rather than mechanically through a passive linkage.

Cooperative vs. autonomous control. The SHR belongs to the cooperative (hands-on) control modality, analogous to the Acrobot. The key difference is the type of VF: the Acrobot uses a forbidden-region VF to prevent the cutter from crossing a planned bone boundary; the SHR uses a guidance VF to attract the needle tip toward a planned path or target point. Both exploit the admittance framework: force in, velocity out.

3.2. Overall Control Law of the SHR

Given a wrench $\mathbf{f} \in \mathbb{R}^6$ (force and torque) measured by the force/torque sensor at the handle, the SHR computes a **tool velocity screw**:

$$\mathbf{v} = \mathbf{G}\mathbf{f}$$

where $\mathbf{G} \in \mathbb{R}^{6 \times 6}$ is the admittance gain matrix that shapes the motion response to force inputs. The specific structure of $\mathbf{G}$ implements the guidance virtual fixture and is derived in Section 4.

The RCM constraint is enforced simultaneously by formulating the problem as a **constrained optimisation**. At each control step, the joint displacement $\Delta\mathbf{q}$ (proportional to joint velocity) is found by minimising:

$$\|\mathbf{J}_h(\mathbf{q})\Delta\mathbf{q} - \mathbf{G}\mathbf{f}\|$$

subject to the constraint:

$$\|\mathbf{P}_{cl} + \mathbf{J}_{cl}(\mathbf{q})\Delta\mathbf{q} - \mathbf{P}_O\| < \varepsilon$$

where:

- $\mathbf{J}_h(\mathbf{q})$ is the manipulator Jacobian resolved at the handle (where the surgeon applies force),
- $\mathbf{J}_{cl}(\mathbf{q})$ is the Jacobian resolved at $\mathbf{P}_{cl}$,
- $\mathbf{P}_{cl}$ is the point on the robot tool currently closest to the virtual RCM point $\mathbf{P}_O$ (the scleral entry point),
- ε is a tolerance on the RCM constraint.

In words: find the joint velocities that realise the Cartesian velocity dictated by the guidance VF, while ensuring that the point on the tool closest to the RCM does not move away from the entry point by more than ε . The objective function minimises the error between the achieved handle velocity and the desired velocity $\mathbf{G}\mathbf{f}$; the constraint enforces the RCM condition with a prescribed tolerance. The problem is treated in discrete time, so $\Delta\mathbf{q}$ represents the joint displacement over one time step, but dividing by the time step recovers the joint velocity interpretation.

4. The GVF Admittance Law

(note: the notation is different from slides)

4.1. General Formulation

The admittance matrix $\mathbf{G}$ is not chosen arbitrarily: it is constructed from the geometry of the planned surgical task to implement a guidance virtual fixture. Two elementary task geometries are considered, and they can be combined for more complex procedures:

1. **Reference target:** the tool tip should move toward a fixed target point in space.
2. **Reference path:** the tool tip should follow a planned curve in space.

The general construction proceeds as follows. Let $\mathbf{P}(t) \in \mathbb{R}^{3 \times k}$ ($0 < k < 3$) be a time-varying matrix whose k columns span the **instantaneous preferred directions of motion** for the tool tip:

- $k = 1$: motion is guided along a curve in space (one preferred direction: the tangent to the curve).
- $k = 2$: motion is guided on a surface (two preferred directions: the tangent plane to the surface).

Define the **orthogonal projection operator** onto the subspace spanned by $\mathbf{P}$:

$$\mathbf{\Pi}_P = \mathbf{P}(\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T$$

and decompose the applied force $\mathbf{f}$ into its preferred and non-preferred components:

$$\mathbf{f}_P = \mathbf{\Pi}_P \mathbf{f}, \quad \mathbf{f}_\perp = (\mathbf{I} - \mathbf{\Pi}_P) \mathbf{f}, \quad \mathbf{f}_\perp^T \mathbf{f}_P = 0$$

The admittance law is then:

$$\mathbf{v} = c(\mathbf{f}_P + \lambda_\perp \mathbf{f}_\perp) = c(\mathbf{\Pi}_P + \lambda_\perp (\mathbf{I} - \mathbf{\Pi}_P)) \mathbf{f} \quad (1)$$

where $c > 0$ is the isotropic admittance gain and $\lambda_\perp \in [0, 1]$ is the **attenuation factor** for the non-preferred force components:

$\lambda_\perp$	Effect
0	Hard GVF: non-preferred force is fully suppressed; the tool can only move in the preferred direction(s).
(0, 1)	Soft GVF: non-preferred force is attenuated but not eliminated; the tool can deviate from the preferred direction with reduced speed.
1	Isotropic admittance: the VF has no effect; all directions are treated equally.

Equation (1) is the general form of an **admittance control law with a time-varying, direction-dependent gain**. The guidance VF is entirely encoded in the projection operator $\mathbf{\Pi}_P$, which changes as the planned path evolves or as the tool tip moves along it.

4.2. Reference Target GVF

When the task is to guide the tip to a **target point** $\mathbf{p}^* \in \mathbb{R}^3$, the preferred direction at the current tip position $\mathbf{p}_t$ is the unit vector pointing from $\mathbf{p}_t$ to $\mathbf{p}^*$:

$$\mathbf{u}(\mathbf{p}_t) = \frac{\mathbf{p}^* - \mathbf{p}_t}{\|\mathbf{p}^* - \mathbf{p}_t\|}$$

Setting $\mathbf{P} = \mathbf{u}$ (a single preferred direction, $k = 1$) and substituting into (1) gives the reference target admittance law. The tip is attracted toward $\mathbf{p}^*$: any force component aligned with $\mathbf{u}$ is transmitted at full gain c , while transverse components are attenuated by $\lambda_\perp$.

Singularity at the target. The matrix $\mathbf{P} = \mathbf{u}$ loses rank as $\mathbf{p}_t \rightarrow \mathbf{p}^*$ (the preferred direction becomes undefined). The standard remedy is to define a **switching sphere** $\mathcal{S}(\delta)$ centred on $\mathbf{p}^*$ with radius δ . Outside $\mathcal{S}(\delta)$, the VF is active with the moving preferred direction. Inside $\mathcal{S}(\delta)$, one of two strategies is adopted:

1. Fix the preferred direction to its last computed value and continue with the VF, or
2. Switch to a simple autonomous proportional control: $\mathbf{v} = \alpha(\mathbf{p}^* - \mathbf{p}_t)$.

Velocity discontinuities on the switching surface can be avoided by using a VF-based *acceleration* control scheme instead of a velocity scheme.

4.3. Reference Path GVF

When the task is to guide the tip along a **planned curve** $\gamma(s)$ (parametrised by arc length $s \in [0, L]$), the construction is more involved because a simple substitution of $\mathbf{P} = \mathbf{t}$ (the tangent to the curve at the closest point) would not recover an initial offset between the tip and the curve.

The construction proceeds as follows:

1. Find the parameter value $\hat{s}(\mathbf{p}_t)$ such that $\gamma(\hat{s})$ is the point on the curve closest to the current tip position:

$$\hat{s}(\mathbf{p}_t) = \arg \min_{s \in [0, L]} \|\gamma(s) - \mathbf{p}_t\|$$

2. Compute the unit tangent to the curve at the closest point:

$$\mathbf{t}(\hat{s}) = \frac{\gamma'(\hat{s})}{\|\gamma'(\hat{s})\|}$$

3. Compute the **lateral error** (the vector from the tip to the closest point on the curve):

$$\mathbf{e}(\mathbf{p}_t) = \gamma(\hat{s}(\mathbf{p}_t)) - \mathbf{p}_t$$

4. Define the **corrected preferred direction** that combines the tangent with a correction term pulling the tip back onto the curve:

$$\tilde{\mathbf{u}}(\mathbf{p}_t) = \text{sign}(\mathbf{f} \cdot \mathbf{t}(\hat{s})) \mathbf{t}(\hat{s}) + k_e \mathbf{e}(\mathbf{p}_t)$$

where $k_e > 0$ is a design gain that weights the error correction against the forward progress along the curve.

The resulting preferred direction $\tilde{\mathbf{u}}$ is not purely tangential: it has a component that drives the tip transversally back onto the curve whenever there is a lateral offset. Substituting $\mathbf{P} = \tilde{\mathbf{u}}$ into (1) gives the reference path admittance law.

4.4. Convergence and Practical Extensions

Under the assumption that $\mathbf{P}$ is full column rank (satisfied everywhere except at the target singularity) and that the human input force is continuous, the admittance law (1) is guaranteed to make the tip converge **asymptotically** to the preferred subspace (the target point or the reference curve). For the path case, the lateral error $\mathbf{e}$ decays to zero, and the tip converges to the curve while advancing along it in the direction of the applied force.

In practice it is useful to extend the basic framework to confine the tool tip not just to a curve or target, but to a **tube** around the path (a cylinder of radius r about the curve) or to a **cone** with vertex at the target. These extensions are obtained by combining the reference target and reference path GVFs and by modifying the switching conditions.

5. Similar Systems for Ophthalmic Surgery

Several research groups worldwide have developed robotic systems for eye surgery that implement analogous GVF-based cooperative control. These are noteworthy because they demonstrate the generality of the approach across different kinematic architectures and institutions.

University Hospitals Leuven. First *in vivo* clinical application of a robot-assisted retinal surgery system was performed at University Hospitals Leuven. The robot developed there uses an RCM principle implemented mechanically via a parallel linkage, combined with a cooperative control law. The system has been used for sub-retinal injections in humans, demonstrating clinical feasibility.

TU Munich. The group at the Technical University of Munich developed a hybrid parallel-serial robot specifically designed for ophthalmic surgery. Virtual fixture control is applied to the end-effector to enforce the RCM constraint and to guide the surgical instrument. The hybrid architecture aims to combine the stiffness advantages of parallel mechanisms near the eye with the larger workspace of serial robots.

Micron (Carnegie Mellon University). Micron is an **active handheld micromanipulator**: unlike the SHR or the Leuven/TU Munich systems, it does not share mechanical support of the tool with a robot arm. Instead, it is a self-contained handheld device equipped with high-bandwidth piezoelectric actuators. Position measurements (rather than forces applied to a handle) are the primary input, and the virtual fixtures are generated in real time from **microscope video** during the procedure. Micron directly cancels the surgeon's tremor by measuring it and commanding compensatory actuator motion, while VFs generated from the live microscope image constrain the tool to remain on a planned path or within a safe zone.

A common principle across diverse architectures. The SHR, the Leuven robot, the TU Munich system, and Micron all address the same clinical problem (tremor and scale mismatch in microsurgery) with different hardware strategies. Yet all of them ultimately implement some form of direction-dependent attenuation of the surgeon's motion — whether through an admittance matrix, an active tremor cancellation loop, or an image-based VF. The underlying mathematical framework introduced in Lectures 8 and 9 applies to all of them.

6. Impedance-Based Dynamic Virtual Fixtures in Teleoperated Systems

6.1. Setup: da Vinci Research Kit

The second major topic of this lecture concerns VFs in **teleoperated** (as opposed to cooperative) systems, specifically the **da Vinci Research Kit (dVRK)**. In this setting, the surgeon operates a **Master Tool Manipulator (MTM)** and the robot executes the corresponding motion at the remote (slave) end. Virtual fixtures are applied to the MTM using **impedance-based control**.

The impedance dynamics of the MTM in the task space, achieved through feedback control, are written as:

$$\mathbf{M}(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}) + \mathbf{D}(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}) = \mathbf{f}_h + \mathbf{f}_{VF} \quad (2)$$

where $\mathbf{x}$ and $\mathbf{x}_d$ are the actual and desired task-space positions of the MTM end-effector, $\mathbf{M}$ and $\mathbf{D}$ are the apparent inertia and damping matrices of the MTM, $\mathbf{f}_h$ is the force applied by the human operator, and $\mathbf{f}_{VF}$ is the virtual fixture force to be designed.

6.2. GVF Constraint Enforcement via Spring-Damper Force

The constraining path for the GVF is defined using the methods described in Section 4 (or analogous geometric planning). Once the path is defined and the closest point $\mathbf{x}_d$ on the path to the current MTM position $\mathbf{x}$ is identified, the GVF must exhibit **attractive behaviour** toward the path. This is realised by a **spring-damper force**:

$$\mathbf{f}_{VF} = -\mathbf{K}_{VF}(\mathbf{x}_d - \mathbf{x}) - \mathbf{D}_{VF}(\dot{\mathbf{x}}_d - \dot{\mathbf{x}})$$

where $\mathbf{K}_{VF} > 0$ is the VF stiffness matrix and $\mathbf{D}_{VF} > 0$ is the VF damping matrix. Both are in general anisotropic: higher stiffness and damping in the non-preferred directions enforce hard guidance, while reduced values in those directions give soft guidance.

Substituting into (2) gives the **closed-loop dynamics**:

$$\mathbf{M}(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}) + (\mathbf{D} + \mathbf{D}_{VF})(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}) + \mathbf{K}_{VF}(\mathbf{x}_d - \mathbf{x}) = \mathbf{f}_h$$

This is a mass-spring-damper interaction law between the human force $\mathbf{f}_h$ and the position error relative to the path. The spring $\mathbf{K}_{VF}$ drives the MTM back to the path when the operator deviates; the damping $\mathbf{D} + \mathbf{D}_{VF}$ dissipates oscillations.

6.3. Passivity and Dynamic Adaptation

When the VF parameters ($\mathbf{K}_{VF}$, $\mathbf{D}_{VF}$) are adapted online — for example, because the constraining path is updated in real time from intra-operative sensing — it becomes necessary to **render the closed-loop system passive** with respect to the input-output pair $(\mathbf{f}_h, \dot{\mathbf{x}}_d - \dot{\mathbf{x}})$.

Passivity is a stability property: a passive system cannot generate energy internally, so it is guaranteed not to produce unbounded oscillations regardless of how the environment (here, the human operator) behaves. Ensuring passivity is particularly important when VF parameters change dynamically, because such changes can inject energy into the system and destabilise the interaction. Passivity-preserving adaptation strategies (such as energy tanks or time-domain passivity control) are used to guarantee stability while allowing real-time VF reconfiguration.

Master-side vs. slave-side VF. Applying the VF force $\mathbf{f}_{VF}$ to the master device (as described here) means the operator directly *feels* the constraint as a haptic resistance. This provides the most informative feedback. However, it does not prevent the operator from overriding the constraint by applying sufficient force. For safety-critical applications, a slave-side enforcement is preferred (the slave robot physically resists entering the forbidden region), and both sides can be combined.

7. Summary and Key Conceptual Threads

- **GVFs are direction-selective admittances.** The anisotropic admittance matrix $\mathbf{G}$ in the SHR control law is entirely determined by the projection operator $\mathbf{\Pi}_P$ onto the preferred direction subspace. Changing the task geometry (target vs. path, hard vs. soft) changes only $\mathbf{P}$ and $\lambda_{\perp}$; the controller structure remains the same.
- **The RCM constraint is enforced as an optimisation constraint.** Rather than mechanically imposing the RCM, the SHR adds it as a constraint to the joint velocity optimisation problem. This is a general strategy applicable whenever a kinematic constraint must be satisfied simultaneously with a VF-based admittance law.
- **Impedance-based dynamic VFs use spring-damper forces.** In the dVRK context, the GVF is realised by adding a spring-damper force $\mathbf{f}_{VF}$ to the master manipulator's impedance dynamics. The resulting closed-loop is a mass-spring-damper; the spring drives the tool to the path and the damper prevents oscillation.

- **Passivity is essential for stability under dynamic VF adaptation.** When VF parameters change online, energy can be injected into the system. Passivity-based methods (energy tanks, time-domain passivity control) guarantee stability without restricting the admissible parameter changes.

Key topics for revision.

1. Clinical motivation for robotic assistance in retinal vein cannulation: scale of the task vs. amplitude of hand tremor.
2. The SHR control law: the admittance $\mathbf{v} = \mathbf{G}\mathbf{f}$, the constrained optimisation for enforcing the virtual RCM, and the meaning of $\mathbf{P}_{cl}$, $\mathbf{P}_O$, $\mathbf{J}_h$, $\mathbf{J}_{cl}$.
3. The general GVF admittance law (Eq. 1): the projection operator $\mathbf{\Pi}_P$, the attenuation factor $\lambda_{\perp}$, and how they implement hard/soft guidance.
4. Reference target GVF: preferred direction, the singularity at the target, and the switching sphere remedy.
5. Reference path GVF: closest point on the curve, the tangent direction, and the correction term for lateral error.
6. Impedance-based dynamic VFs in the dVRK: the spring-damper force $\mathbf{f}_{VF}$, the closed-loop mass-spring-damper dynamics, and why passivity matters under dynamic adaptation.

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